
Data Structures

BS (CS) _Fall_2024

Lab_04 Tasks



Learning Objectives:

1. Sorting Algorithms
2. Searching Algorithms

Lab Tasks

Submission Instructions

1. Create a separate **.h** and **.cpp** file for each class.
2. Now create a new folder with name *ROLLNO_SEC_LAB01* e.g. **i23XXXX_A_LAB01**(Also name your **cpp** file like this.)
3. Move all of your files to this newly created directory and compress it into a **.zip file**. Return the test file as well.
4. Now you have to submit this zipped file on Google Classroom.
5. If you don't follow the above-mentioned submission instruction, you will be marked **zero**.
6. Plagiarism or AI generated code in the Lab Task will result in **zero** marks in the whole category.

Q 1. Eric loves puzzle making. One night he went to sleep after completing his puzzle only to wakeup and find his puzzle had been messed up. Help Eric in **sorting** the puzzle pieces back to correct form.

- ☐ The first row of puzzle has to be sorted using **selection sort**
- ☐ The second row of puzzle has to be sorted using

bubble sortPrototype:

Unsorted Puzzle

32	30	39	34	37	35	33					
50	48	54	59	47	49	52	51	55	53	57	58

Sorted Puzzle

30	32	33	34	35	37	39					
47	48	49	50	51	52	53	54	55	57	58	59

Q2. Problem Statement: You are given a sentence S consisting of lowercase English alphabets and spaces. Your task is to find all the vowels in the sentence S using binary search algorithm. and construct a new string X containing only these vowels. After constructing the string X, you need to answer several queries to determine how many times a specified vowel appears in the original sentence S.

Input Format

- The first line contains a sentence S consisting of lowercase English letters and spaces.
- The second line contains an integer Q, representing the number of queries.
- The third line contains Q space-separated characters, where each character is a vowel (a, e, i, o, u).

Output Format

For each query, print a line in the following format:

"<vowel> appears <count> times"

where <vowel> is the vowel from the query, and <count> is the number of times it appears in the original sentence S.

Example

Sample Input	Sample Output
this is my lab task query.	a appears 2 times
5	e appears 1 times
a e i o u	i appears 2 times
	o appears 0 times
	u appears 1 times

Q3. Write a C++ function to find the position where a target integer x would be inserted in a sorted array such that the array remains sorted using binary search.

Prototype: int findInsertPosition(int arr[], int n, int x);

Q4. Write a C++ function to perform a linear search for a target string in a list of strings. The function should return the index of the target string if found, otherwise return -1.

Prototype: int linearSearchString(string arr[], int n, string target);